

Unit

2

Lesson

1



Innovators in Action.

Builders at Work!

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Describe how homes can be changed to help people with disabilities.
- Use materials like cardboard, clay, and simple shapes to build furniture and walls.
- Create simple patterns for decoration in room models.
- Write and compare 2- and 3-digit numbers (e.g., room sizes).
- Use place value knowledge to explain the meaning of each digit in a number.

Challenge questions of the lesson

1. Can you explain why ramps, wide doors, or grab bars help people move safely and easily?
2. How can working together as a team help us build a better accessible home model?

SEL Relation

- **Self-Awareness:** Students reflect on their feelings about accessibility and how it impacts people's daily lives, fostering empathy and personal connection.
- **Self-Management:** Through hands-on building and teamwork, students practice patience, focus, and perseverance while solving design challenges.
- **Social Awareness & Relationship Skills:** Listening to stories and collaborating in groups helps students understand diverse needs, communicate effectively, and work cooperatively.
- **Responsible Decision-Making:** Students make thoughtful, ethical choices in designing accessible features, explain their reasoning, and use feedback to improve their work.

Engage



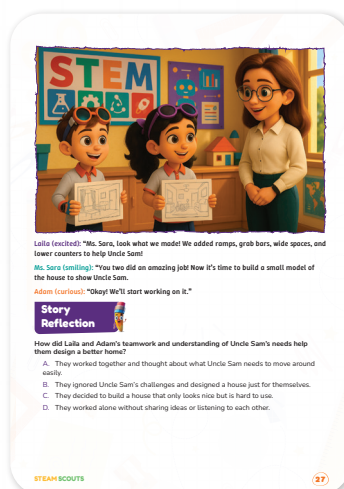
Teacher's Role

- The teacher's role is to set an engaging and curious atmosphere, read the story aloud with expression, and facilitate initial student reactions and reflections. The teacher will use strategies to spark interest and encourage an interactive and reflective classroom environment.

Expected From Students:

- Students are expected to actively listen to the story, react to its content, and reflect on the main message, key lesson, or idea. They should be prepared to connect the story's themes to their own experiences.

Parts of the books included.



Story



1. Set the Scene with Curiosity and Care

Strategy: Think-Pair-Share (Kagan Structure). Begin by asking students what they know about making homes comfortable for everyone, especially people who might need special help to move around. For instance, "What kind of things do we have in our homes that help us? What if someone used a wheelchair, what might be hard for them in a house?"

2. Read the Story Aloud with Expression

Pose the question to the class. Give students individual "think time" to consider their answers. Then, have them "pair" with a partner to discuss their ideas. Finally, "share" some of their ideas with the whole class, encouraging a diverse range of responses.

Possible Strategies for Implementation:

- **Partner Discussion (Kagan Strategy):** After reading the story, have students pair up and talk about this prompt: “What was the most helpful thing Lila and Adam did for Uncle Sam?”
- **Create a Headline (Thinking Routine):** Invite students to write a short headline, like one from a newspaper, that captures the main message of the story. This helps them focus on the key idea.
- **See–Think–Wonder (Thinking Routine):** Show students a picture of an accessible home feature (such as a ramp or grab bar) after the story. Ask them: “What do you notice? What do you think it’s for? What questions do you have about it?”

Story Reflection



Objective:

- Students will be able to identify how teamwork and understanding individual needs lead to effective solutions.
- Students will be able to reflect on the core message of the story regarding empathy and problem-solving.

Answers:

Story Reflection



How did Lila and Adam’s teamwork and understanding of Uncle Sam’s needs help them design a better home?

- A. They worked together and thought about what Uncle Sam needs to move around easily.
- B. They ignored Uncle Sam’s challenges and designed a house just for themselves.
- C. They decided to build a house that only looks nice but is hard to use.
- D. They worked alone without sharing ideas or listening to each other.

Possible Strategies for Implementation:

Present the Story Reflection Question: Show students the question: “In what ways did Lila and Adam’s teamwork and awareness of Uncle Sam’s needs help them improve their home design?”

- **Strategy:** Use the Think-Pair-Share approach (Kagan Structure).
- **Implementation Steps:** Ask students to first reflect quietly on the question. Then, have them discuss their ideas with a partner. Finally, invite several pairs to share their responses with the class. As they share, steer the discussion toward answer A as the most complete and thoughtful response.

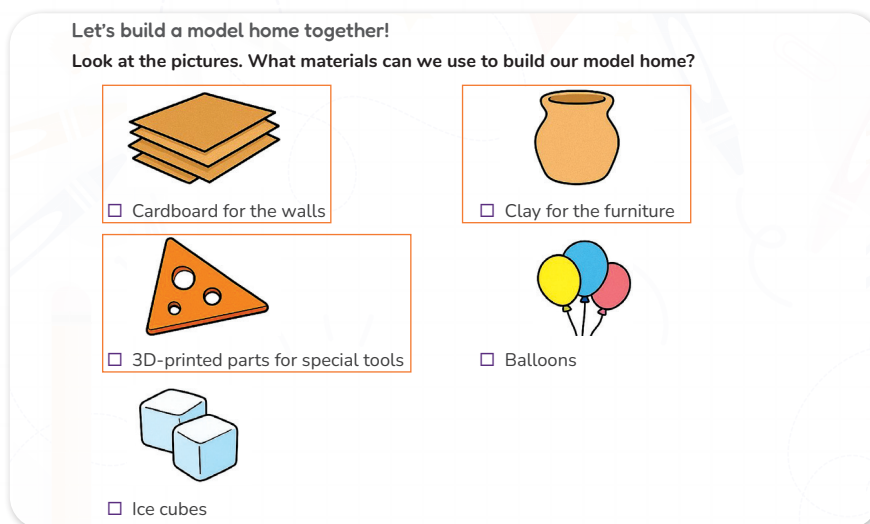
Let's Think!



Objective:

- Students will be able to identify suitable materials for building a model home.
- Students will be able to reflect on the core message of the story regarding empathy and problem-solving

Answers:



Possible Strategies for Implementation:

Lead the “Let’s Think” Exploration with Materials and Inquiry Questioning

Strategy Used: Combine See-Think-Wonder (Visible Thinking Routine) with Brainstorming and Sorting (Inquiry-Based Learning).

• How to Do It:

- Begin by showing the “Let’s Build a Model Home Together!” image. Give students a moment to quietly think about what they notice.
- Use the See-Think-Wonder questions to guide discussion: “What do you see in the picture? What do you think about these materials? What do you wonder about using them to build a model home?”
- Center the conversation around the guiding question: “Which materials would be good for building our model home?”
- In small groups, have students discuss the options (Cardboard, Clay, 3D-printed parts, Balloons, Ice cubes). Ask them to choose which materials they would use and explain their reasoning.
- Encourage the class to develop deeper questions, such as: “How can we design homes to be easier for everyone to use?” or “How can we make houses more accessible for people with wheelchairs using materials like these?”
- Write their inquiry questions on chart paper to revisit throughout the unit.



Teacher's Role:

- The teacher's role is to facilitate the "Screen Lesson" by setting a purpose for watching, guiding active engagement during the video, and leading a discussion and reflection afterward. The teacher will connect the video content back to the students' DIY lab experience and reinforce key concepts.

Expected From Students:

- Students are expected to actively watch the video, relate its content to their DIY lab projects, and participate in discussions to deepen their understanding. They should be able to articulate what they learned and how it connects to their hands-on experience.

Parts of the books included.



Screen Lessons

Click on the link to watch the video and learn more about place value.

https://www.youtube.com/watch?v=_dHu5TFxPtk



Screen Lessons

Objective:

Students will be able to understand the concept of place value and its application in real-world contexts like room measurements, and relate it to designing functional spaces.

Video link: https://www.youtube.com/watch?v=_dHu5TFxPtk



Possible Strategies for Implementation:

1. Before Watching – Set a Purpose

- **Approach:** Use a KWL Chart (Visible Thinking Routine).
- **Implementation:** Draw a KWL chart on the board. Ask students what they already know about numbers and measuring spaces, and what they want to learn about place value or designing accessible rooms. This helps connect to their prior knowledge and gives them a purpose for watching the video.

2. While Watching – Stay Engaged

- **Approach:** Pause and Jot (Adapted from Kagan Strategy).
- **Implementation:** Play the video and pause at important moments, especially when explaining place value or space measurement. After each pause, students write down a new idea, a question, or how it connects to their model home project. For example, pause when the 'tens' place is explained and ask, "How does understanding tens help us think about kitchen space?"

3. After Watching – Reflect and Discuss

- **Approach:** Combine Think-Pair-Share with the "Connect, Extend, Challenge" thinking routine.
- **Implementation:** Have students think about what the video taught them, discuss it with a partner, and then share with the class. Guide them to reflect on how the content connects to what they already knew, what new ideas it added, and what questions or challenges it raised.

Elaborate and Evaluate



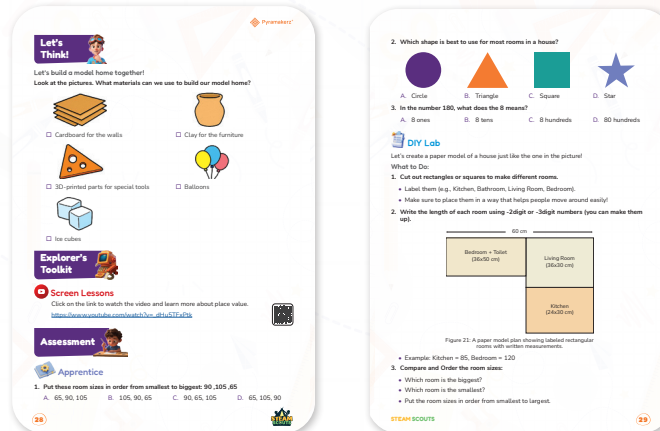
Teacher's role

- **Facilitate Assessment:** The teacher's primary role is to administer and guide students through the "Apprentice" assessment. This includes presenting the multiple-choice questions (MCQs) and guiding access to any game-based assessment.
- **Monitor Understanding:** Observe students as they answer questions and engage with the assessment. This helps gauge their individual and collective understanding of the video content and concepts reinforced during the "Screen Lesson" and DIY lab.
- **Provide Support (as needed):** While it's an assessment, the teacher can offer clarification on instructions or questions, ensuring students understand what is being asked, without giving away answers.
- **Encourage Critical Thinking:** Even during assessment, the teacher can encourage students to think carefully about their answers and recall information from previous activities.
- **Connect to Prior Learning:** The teacher should help students connect the assessment questions back to the content of the video related to the DIY lab and other parts of the lesson.

Expected From Students:

- **Answer MCQs:** Students are expected to answer the multiple-choice questions based on the content of the video related to the DIY lab.
- **Engage with Game-Based Assessment:** If a game-based assessment is part of this section, students are expected to actively participate and demonstrate their understanding through the game.
- **Demonstrate Understanding:** Students should show their comprehension of the key concepts and information presented in the "Screen Lesson" video and reinforced by the DIY lab experience.
- **Apply Knowledge:** Students should apply the knowledge gained from the video and previous activities to correctly answer the assessment questions. Here's the teacher's role and expected student outcomes for the "Apprentice"

Parts of the books included.



Apprentice

Objective:

Students will be able to understand the concept of place value and its application in real-world contexts like room measurements, and relate it to designing functional spaces.

Answers:

Put these room sizes in order from smallest to biggest: 90 ,105 ,65

A. 65, 90, 105

Which shape is best to use for most rooms in a house?

C. Square

In the number 180, what does the 8 mean?

B. 8 tens

Activity instructions:

1. **Think on Your Own:** After watching the video about place value and measuring rooms, take some time to read through the “Apprentice” questions and reflect on your answers.
2. **Talk with a Partner:** Pair up with a classmate to compare your answers and explain your thinking.
3. **Share with the Class:** Join the group discussion and share your ideas and reflections with everyone.

Possible Strategies for Implementation:

- **Strategy Used:** Numbered Heads Together (Kagan Structure)
- **How to Use It:** In small groups, give each student a number. Ask either the “Apprentice” multiple-choice questions from the “Explain” section or new ones based on the “Builder” video (if applicable). Students collaborate to make sure everyone in the group understands the answer. Then, call out a number, and the student with that number from each group will explain their group’s response. This approach helps reinforce key ideas through teamwork and accountability.

Explorer's Toolkit



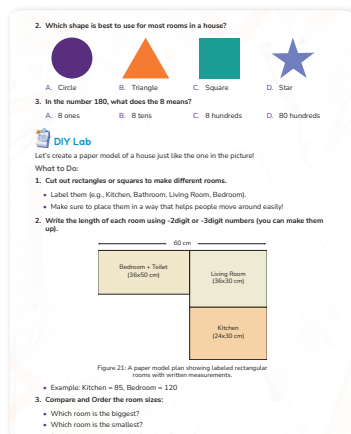
Teacher's Role:

- Guide students in brainstorming ideas for their house models, facilitate collaborative discussions, and provide support and troubleshooting as students design and build their simple machines or accessible features. The teacher will encourage trial and error and foster a creative, problem-solving environment.

Expected From Students:

- Students will collaboratively design and build a simple paper model of a house with three rooms (bedroom, kitchen, bathroom). They will label the rooms, assign them lengths using 2-digit or 3-digit numbers, and then compare and order these room sizes. They will also begin to think about how to make these rooms accessible.

Parts of the books included



DIY Lab

Objective:

Students will apply spatial reasoning to design a functional house layout, using their understanding of place value and number comparison with 2- and 3-digit numbers. They will also begin to incorporate accessible design features into their model homes, encouraging inclusive thinking. This activity reinforces mathematical reasoning, design planning, and awareness of diverse user needs.

Materials Needed

- Paper or Cardboard
- Scissors

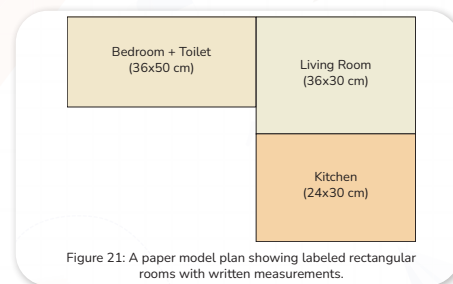
Activity Instructions:

Let's create a paper model of a house just like the one in the picture!

1. Cut out rectangles or squares to make different rooms.

- Label them (e.g., Kitchen, Bathroom, Living Room, Bedroom).

- Make sure to place them in a way that helps people move around easily!
2. **Write the length of each room using -2digit or -3digit numbers (you can make them up).**
- Example: Kitchen = 85, Bedroom = 120
3. **Compare and Order the room sizes:**
- Which room is the biggest?
 - Which room is the smallest?
 - Put the room sizes in order from smallest to largest.



Possible Strategies for Implementation:

- **RallyCoach (Kagan Structure):** Pair students to work on designing and measuring the rooms. One student explains and supports while the other completes the task, then they switch roles. This builds confidence and encourages cooperative learning.
- **Think-Pair-Share (Kagan Structure):** Before students begin cutting their room models, have them think individually, then discuss with a partner how to arrange the rooms to support easy movement—especially with Uncle Sam’s needs in mind.
- **Measure Up (Visible Thinking Routine):** While assigning and comparing room sizes, prompt students with reflective questions like: “What’s large about this design? What’s small? What stands out? What matters most?” to help them evaluate their spatial choices.
- **Chalk Talk (Visible Thinking Routine):** If students need help thinking about accessibility, use a silent brainstorming method. Post a prompt such as, “What changes would make this kitchen easier for Uncle Sam to use?” and allow students to draw or write ideas quietly in a shared space, building off one another’s thoughts.

Elaborate and Evaluate



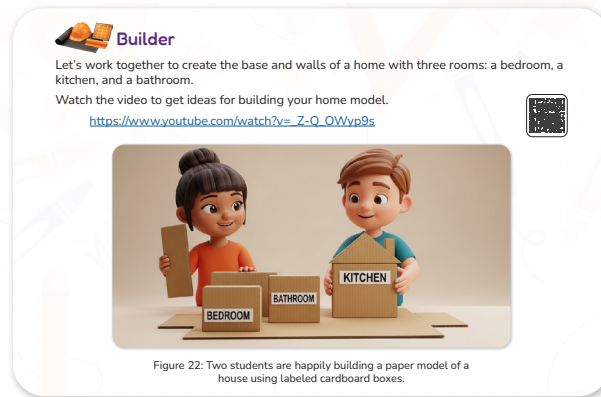
Teacher’s Role:

- Guide students as they move from building the basic structure of their model homes to designing and adding furniture.
- Support students in ensuring their furniture choices match their plans for accessibility.
- Encourage critical thinking about how each furniture piece can help make the home easier to use for all people.

Expected from Students:

- Plan and begin constructing furniture pieces for their model homes.
- Make thoughtful design choices that reflect the accessibility goals in their original plans.
- Consider how each item supports easier movement and use within the space.

Parts of the books included.



Builder

Objective:

Students will be able to translate their overall house design plans into the concrete creation of furniture, focusing on how each piece contributes to making the model home more accessible and functional.

Resources: https://www.youtube.com/watch?v=_Z-Q_OWyp9s



Materials Needed:

- Cardboard sheets or sturdy cardstock
- Scissors or child-safe cutters
- Ruler and pencil for measuring
- Markers or labels (to name the rooms)
- Glue, glue sticks, or masking tape
- Printable room labels (optional)
- Template or sample video for guidance

Activity Instructions:

- **Watch and Observe:** Begin by watching the linked video to understand how to build the base and walls.
- **Draw and Cut:** Use a ruler and pencil to outline three room walls (bedroom, kitchen, and bathroom) and a rectangular base on cardboard.
- **Label the Rooms:** Clearly label each piece before assembly to ensure correct placement.
- **Assemble the Model:** Use glue or tape to attach the walls to the base, forming the 3-room layout as shown in the image.
- **Check for Stability:** Reinforce with extra tape or tabs if needed to keep the walls upright and stable.
- **Optional:** Begin thinking about furniture or accessibility features for the next step in the project.

Possible Strategies for Implementation:

1. Watch-Think-Do (Direct Instruction + Application):

- Start by showing the video, then pause to discuss each step before students begin building.

2. Step-by-Step Modeling:

- The teacher models each step while students follow along, checking after each step for understanding.

3. Peer Collaboration (Kagan RallyCoach):

- Students work in pairs, taking turns building and coaching each other through the steps.

4. Think-Pair-Share (Before Building):

- Ask students to discuss with a partner how they might arrange the rooms and why (especially in terms of easy movement or accessibility).

5. Visual Reference Wall:

- Post labeled diagrams or steps on the board to help visual learners as they build.

6. Design Challenge Prompt (Extension):

- Ask: “How would you change your layout to make it easier for Uncle Sam to move around?”

Elaborate and Evaluate



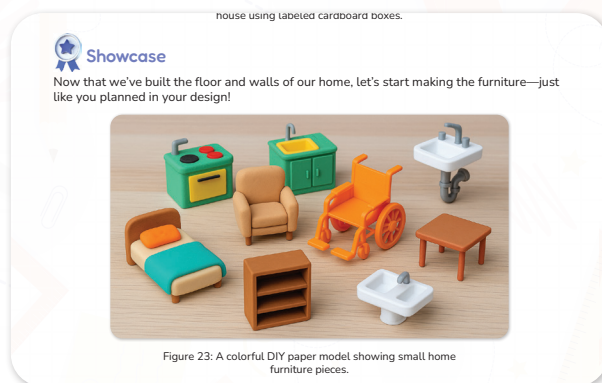
Teacher's Role

- Facilitate and guide students as they transition from structure building to furnishing their homes.
- Encourage students to reflect on their design plans and ensure furniture choices align with accessibility goals.
- Support students in using different materials creatively to solve design challenges.
- Prompt students to explain how each furniture item helps someone like Uncle Sam move around and use the space comfortably.

Expected from Students

- Build furniture that matches their earlier accessibility design.
- Explain how each piece of furniture supports movement and usability.
- Collaborate, test placements within rooms, and make improvements as needed.
- Participate in a class showcase or walkthrough where they present their furnished model homes.

Parts of the books included



Objective:

Students will design and create furniture pieces for their model homes based on their accessibility plans. This hands-on activity helps reinforce spatial reasoning, planning skills, and inclusive design thinking while allowing students to showcase their creativity and problem-solving.

Materials Needed

- Recycled materials: bottle caps, small boxes, straws, fabric scraps, buttons
- Modeling clay or air-dry clay
- Cardboard scraps or foam board
- Glue, tape, scissors
- Markers or paints
- Optional: pre-made mini accessories or 3D-printed parts (wheelchair, sink, etc.)

Activity Instructions

1. Review your room layout and original design plans.
2. Decide what furniture is needed in each room (e.g., bed, sink, stove, table).
3. Use available materials to build each furniture item, keeping accessibility in mind.
4. Test the size and placement inside your model home.
5. Make adjustments to ensure there's enough space for movement.
6. Prepare to share your finished model and explain how it supports Uncle Sam's need

Possible Strategies for Implementation

- RallyCoach (Kagan Structure): Students work in pairs—one builds while the other gives feedback, then switch roles. This supports collaboration and reflection on design choices.
- Gallery Walk: Set up homes around the room. Have students walk through each other's models and give feedback using sticky notes ("I noticed...", "I wonder...").

- Think Aloud: Ask students to verbally explain their design process while building, reinforcing reasoning and metacognition.
- Design Challenge Reflection: Prompt students with questions like “How does this piece of furniture help someone with mobility needs?” or “What did you need to change from your original plan?”
- Checklist or Rubric-Based Review: Use a simple student-friendly rubric or checklist to evaluate if their furniture meets the design goals (e.g., fits in space, supports access, is stable, etc.).

Evaluation Rubric – Accessible Furniture Showcase.

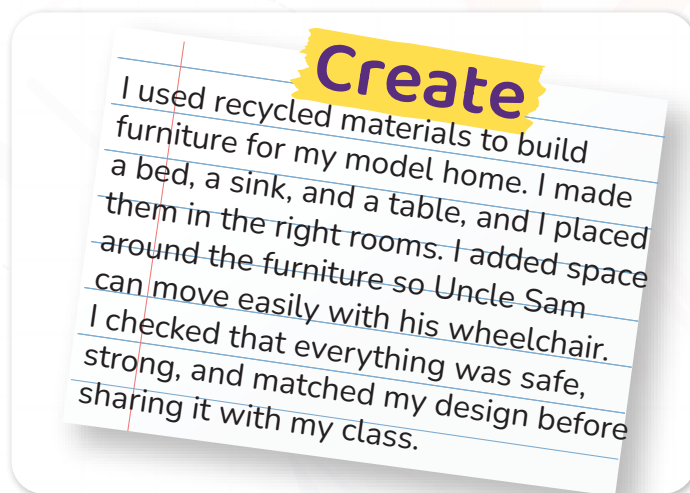
What We're Looking For	Needs Help (1 pt)	Getting There (2 pts)	Great Job! (3 pts)	Wow! Amazing Work! (4 pts)
Did you follow your plan?	Your furniture doesn't look like your plan.	A few things match your plan.	Most furniture matches your plan.	Everything matches what you planned.
Is your home easy to use?	It's hard to move around or use.	Some spaces are easy to use.	Most things are easy to use.	Everything is easy to reach and move around.
Did you build carefully?	Things fall apart or are messy.	Some parts are neat and steady.	Most furniture is neat and strong.	Everything is neat, strong, and well-made.
Did you use materials well?	You didn't use many tools or ideas.	You used some things well.	You used your materials creatively.	You used your tools and materials in a smart and fun way.
Can you explain your choices?	I'm not sure why I built things this way.	I can say a little about what I made.	I can explain most of my choices.	I can clearly explain how my design helps people like Uncle Sam.

Total Score: / 20

How You Did

- 17–20 points: You're a super home designer!
- 13–16 points: Nice job! A few things to work on.
- 9–12 points: You're learning! Keep practicing.
- 5–8 points: Let's try again together!

Don't forget to go back to the EDP page and add these to planning part.



Now I Can

Objective:

Students will reflect on their learning by identifying what they can now do, including building accessible home models, using materials creatively, recognizing patterns, and applying place value skills to compare room sizes. This reflection helps reinforce key math and design concepts and supports metacognitive awareness.

Possible Strategies for Implementation:

- Read Aloud + Thumbs Up: Read each “Now I Can” statement aloud and ask students to give a thumbs up if they feel confident, sideways if they’re unsure, or down if they need more help.
- Check and Chat: Students pick one “Now I Can” statement and talk with a partner about how they showed that skill in the project.
- Sticker Checklists: Give students a copy of the “Now I Can” box. Let them place a sticker or draw a smiley face next to each thing they feel confident doing.
- Reflect and Draw: Ask students to draw or write about one thing they’re proud of from the list (e.g., “I used big numbers to measure the kitchen”).
- Teacher Conference Spot Checks: Briefly check in with students individually or in small groups to ask: “Can you tell me how you made your home easier to use?” or “What number did you use to measure a room?”

Relation to Standard

Mathematics (Common Core State Standards – CCSS)

CCSS.MATH.CONTENT.2.NBT.A.1:

Understand that the three digits of a three-digit number represent amounts of hundreds, tens, and ones.

CCSS.MATH.CONTENT.2.NBT.A.4:

Compare two three-digit numbers using $>$, $=$, and $<$ symbols.

CCSS.MATH.CONTENT.2.MD.A.1:

Measure the length of an object by selecting and using appropriate tools.

Science & Engineering (Next Generation Science Standards – NGSS)

K-2-ETS1-1:

Ask questions, make observations, and gather information about a situation people want to change to define a simple problem.

K-2-ETS1-2:

Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps it function.

Technology & Design (ISTE Standards for Students)

ISTE 4:

Innovative Designer: Students use a variety of technologies within a design process to identify and solve problems.

ISTE 1:

Empowered Learner: Students build knowledge by actively exploring real-world issues and problems.